

Emil Baignan

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EDUCATION

University of California, Berkeley – Haas School of Business

Master of Financial Engineering

Expected January 2028

Berkeley, CA

University of California, Los Angeles

B.S. Mathematics/Economics with Specialization in Computing

September 2022 – June 2026

Los Angeles, CA

Achievements: Cumulative GPA: 3.74/4.0 | Three-time Dean's List recipient

SKILLS & CERTIFICATIONS

Programming: Python (NumPy, pandas, SciPy, statsmodels, scikit-learn, & PyTorch), R, SQL, C++, LaTeX

Mathematics: Machine Learning, Econometrics, Time Series, Optimization, Stochastic Processes, Probability Theory, Linear Algebra, Financial Mathematics, Numerical Analysis

Languages: English (fluent), Thai (fluent), French (intermediate)

PROFESSIONAL EXPERIENCE

California Financial Partners Inc. (LPL Financial)

Glendale, CA

Investments Intern | Advisors: Ira Candib CFP®; Samuel Croucher CFA

July 2025 – September 2025

- Developed liquidity-conscious macro asset pricing systems to analyze term premia and curve steepening, providing quantitative signal generation that guided a 30% fixed-income duration rebalance across HNW portfolios.
- Implemented Python models to compare rate-cut paths and their impact on curve shapes, providing quantitative inputs into scenario heuristics that supported roughly 15-20% reallocation across muni maturities.
- Modeled global cross-sectional equity valuations against FX dollar-weakening vectors to isolate asymmetric upside, directly triggering an 8% tactical capital allocation shift into ex-US equity fund exposure.

RESEARCH EXPERIENCE

University of California, Los Angeles – Anderson School of Management

Los Angeles

MacroStructure Lab | Advisors: Tyler Muir; Valentin Haddad

April 2026 – Present

- Researched high-dimensional ML-SDF estimation frameworks using 46 years of CRSP/Compustat portfolio data.
- Developed a continuous covariance-whitening framework for high-dimensional SDF estimation, showing variance-hierarchy removal collapsed finite-sample ridge OOS pricing R-square (~15%), while preserving OOS Sharpe.
- Synthesized complex statistical results into economically interpretable SDF insights, presenting independent asset-pricing findings to faculty advisors to shape ongoing empirical research.

University of California, Los Angeles – Bruin Reserve Bank

Los Angeles

Quantile Forecasting | QuantConnect Quant League White Paper

November 2025 – January 2026

- Converted academic literature (Park et al., 2022) on IQF-learning architectures into a systematic strategy.
- Implemented constrained numerical monotone spline interpolation to enforce non-crossing quantile curves, reducing tail-instability by 35% under high-volatility regimes.
- Engineered a custom execution framework to resolve root-finding Kelly leverage constraints within the QuantConnect backtesting platform's runtime limits.

University of California, Los Angeles

Los Angeles

YieldCurveForecaster | Project

January 2025 – March 2025

- Built a Dash app to visualize the historical 3D yield curve, forecast methods, and inversion analytics using FRED data.
- Calibrated Nelson-Siegel-Svensson yield-curve models via constrained quasi-Newton optimization with VIX and inflation-adjusted macro features, validating forecasts via walk-forward RMSE, MAPE, R^2 , and directional accuracy.
- Documented findings on legacy term-structure models and exogenous-data heuristic approaches to inform future research directions for forecasting the yield curve as a 3D structure (maturity, time, and macroeconomic regime).

Interests: Hiking/backpacking, traveling, live music, and movies